Design Proposal

YOGYA:
Intelligent Exam Eligibility & Tracking Platform

Submitted By

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1. Problem Statement

India hosts one of the largest and most complex competitive examination ecosystems in the world. Millions of aspirants appear each year for examinations conducted by UPSC, SSC, IBPS, RBI, and state-level Public Service Commissions, each carrying distinct eligibility criteria covering age limits, educational qualifications, social category reservations, permissible attempt counts, and domicile requirements.

Despite this scale, aspirants currently manage eligibility tracking through a fragmented combination of physical marksheets, handwritten notes, bookmarked government websites, and social media groups. No unified platform exists that:

- Automatically reads and stores academic credentials from physical documents using on-device OCR.
- Applies exact eligibility rules of multiple competitive examinations to a candidate's profile in real time.
- Tracks the number of attempts consumed across examinations from a single interface.
- Alerts the candidate to approaching registration deadlines and provides a direct pathway to official portals.
- Projects future eligibility windows based on the candidate's age, category, and academic standing.
- sometimes aspirant fill to keep in mind the number of attempts they are left with

The consequence is twofold: aspirants miss registration deadlines due to poor tracking, and they invest preparation time in examinations for which they are ineligible, discovering this only at the point of application rejection. Yogya is designed to eliminate both failure modes through intelligent automation.

2. Objectives

2.1 Primary Objectives

- Develop a cross-platform mobile application (Android and iOS) using the Flutter framework that serves as an intelligent exam eligibility and attempt-tracking companion for Indian competitive examination aspirants.
- Implement an on-device OCR pipeline capable of extracting structured academic data from photographs of 10th Marksheets, 12th Marksheets, and Graduation Degree / CGPA Certificates with a confidence score of 95 % or above.
- Build a rule-based Eligibility Engine that evaluates a candidate's profile against the exact eligibility criteria of multiple major Indian competitive examinations simultaneously and produces a real-time match percentage per exam.

- Design a personalised Eligibility Timeline that projects the candidate's valid examination windows chronologically, identifies the nearest deadline, and provides a direct link to the official registration portal.
- Provide a secure, privacy-preserving architecture that stores only structured extracted data, purges raw document images immediately after OCR extraction, and encrypts all credentials using hardware-backed storage.

2.2 Secondary Objectives

- Surface a contextual 'Next Steps' card on eligibility confirmation listing additional documents required by each specific examination with links to the official information source.
- Design a comprehensive Profile, Settings, and Help & Support interface allowing aspirants to manage their academic identity, social category, notification preferences, and security settings from one place.
- Establish an extensible Clean Architecture that allows new examinations, new eligibility rules, and new document types to be added without rebuilding the core engine.
- Deliver a brand-consistent user interface with smooth animations, a persistent bottom navigation paradigm, and a hamburger-triggered side drawer for secondary navigation.

3. Target Users & User Personas

3.1 Target User Segments

Yogya is designed for Indian students and young professionals aged 17 to 35 who are actively preparing for one or more competitive examinations.

Segment	Age Range	Primary Exams	Technical Comfort
School Leavers	17–19 yrs	NDA, AFCAT (future)	Moderate - smartphone-native
Graduate Aspirants	21–27 yrs	UPSC CSE, SSC CGL, IBPS PO, RBI Grade B, CDS	High - frequent app users
Multi-Exam Trackers	22–32 yrs	UPSC CSE, SSC, IBPS, State PSC	High - power users

3.2 User Personas

Persona 1 - Aditi Sharma (UPSC Primary Aspirant)

- Age: 23 | Location: Delhi | Category: General | Education: B.A. (Hons.) Political Science, University of Delhi.
- Goal: Clear UPSC Civil Services Prelims; track remaining attempts and upcoming notification deadlines.
- Pain Points: Manually calculating age eligibility against each notification date. Forgetting new UPSC notifications. Unsure whether her 61 % graduation qualifies her for RBI Grade B simultaneously.
- How Yogya Helps: The Eligibility Engine auto-calculates her age against each exam's notification date, shows Attempt #2 of 6 consumed, confirms RBI Grade B eligibility (60 % threshold met), and surfaces the CDS II registration link 14 days before the deadline.

Persona 2 - Rohan Verma (SSC + Banking Multi-Tracker)

- Age: 26 | Location: Lucknow | Category: OBC | Education: B.Com, Lucknow University.
- Goal: Crack SSC CGL or IBPS PO within the next two years, whichever comes first.
- Pain Points: Tracking two different exam cycles simultaneously. Not sure if his OBC NCL certificate needs renewal. Missed IBPS Clerk last year due to a missed notification.
- How Yogya Helps: Live Eligibility Pulse shows SSC CGL at 92 % match and IBPS PO at 85 % match simultaneously. OBC category entered in Profile automatically applies 3-year age relaxation across all calculations. Dashboard flags the next IBPS notification 30 days in advance.

Persona 3 - Arjun Singh (NDA / Defence Aspirant)

- Age: 18 | Location: Chandigarh | Category: General | Education: Class 12 (PCM, 78 %).
- Goal: Join the Indian Navy through NDA. Currently in Class 12.
- Pain Points: Confused about age cutoffs for NDA versus CDS. Unsure whether 78 % PCM qualifies for the Air Force technical stream.
- How Yogya Helps: After scanning his 12th Marksheet (PCM stream auto-detected), the engine shows Eligible for NDA Navy/Army, Eligible for NDA Air Force (78 % ≥ 60 % threshold), and flags his eligibility window at 18 months remaining. AFCAT is shown as future-eligible after graduation.

4. Competitive Analysis

The table below maps Yogya against existing solutions in the Indian exam preparation and document management space. No direct competitor offers the combination of on-device OCR ingestion, rule-based eligibility checking, and attempt tracking in a single mobile application.

Feature	Testbook / Unacademy	DigiLocker	Spreadsheets	Yogya
OCR Document Scanning	No	No	No	Yes - on-device ML
Rule-Based Eligibility Check	No	No	Manual	Yes - automated engine
Attempt Count Tracking	No	No	Manual	Yes - auto from docs
Personalised Deadline Alerts	General alerts	No	Manual	Yes - per-exam, countdown
Registration Portal Links	General links	No	Bookmarks	Yes - contextual on eligibility
Multiple Exam Tracking	Study content	Storage only	Tedious	Yes - core feature
Document Security	Server upload	Govt cloud	Unencrypted	Hardware-encrypted; images purged
Offline Access	Partial	Yes	Yes	Yes - cached data
Category-Aware Engine	No	No	Manual	Yes - Profile screen
Future Eligibility Projection	No	No	No	Yes - timeline predictions

4.1 Key Differentiators

- Only platform combining OCR ingestion + eligibility rule engine + attempt tracking in one mobile application.
- On-device processing ensures raw marksheet images are never transmitted to a server - a strong privacy advantage.
- Contextual registration links appear at the moment of eligibility confirmation, reducing the steps from 'found eligible' to 'application submitted'.
- Social category-aware engine automatically applies category-specific age relaxations and attempt limits without additional user input.

5. Features List

Feature	Description	Screen	Priority
Splash Screen	Branded cold-start screen with JWT check and conditional routing.	Splash	P0 - Critical
Login Screen	Email/password + Google OAuth sign-in with JWT stored in hardware-backed secure storage.	Login	P0 - Critical
10th Marksheet OCR	Extracts board, year, aggregate %, date of birth. Min. 95 % confidence required.	Documents	P0 - Critical
12th Marksheet OCR	Extracts board, stream, aggregate %. Drives NDA / CDS / AFCAT eligibility.	Documents	P0 - Critical
Graduation CGPA / Degree OCR	Extracts university, CGPA (auto-converted via UGC formula), pass year. Drives central govt exam eligibility.	Documents	P0 - Critical
Rule-Based Eligibility Engine	Evaluates age, qualification, category, and attempts simultaneously for 10+ exams.	Eligibility	P0 - Critical
Live Eligibility Pulse	Real-time match % per exam with colour-coded bars and reason text, updates as user fills the wizard.	Eligibility	P0 - Critical
Eligibility Timeline	Chronological exam window view with ELIGIBLE / INELIGIBLE verdict and expand-on-tap detail per node.	Timeline	P0 - Critical
Deadline Alerts	Countdown cards for upcoming registration deadlines with HIGH PRIORITY / EXAM DATE badges.	Dashboard	P0 - Critical
Registration Link Surface	'Register Now' button with official portal link on eligibility confirmation.	Dashboard / Timeline	P0 - Critical
Additional Docs Info Cards	Lists required documents + authoritative info links per exam on eligibility confirmation.	Dashboard / Docs	P0 - Critical
Social Category Selection	Profile-level General / OBC / SC / ST / EWS selector feeding the eligibility engine.	Profile	P0 - Critical

Feature	Description	Screen	Priority
Side Drawer Navigation	Hamburger-triggered overlay with Home, Eligibility Engine, My Documents, Attempt History.	Global	P1 - High
Attempt History Screen	Log of exams attempted, derived from uploaded admit cards and manual entries.	Side Drawer	P1 - High
Future Eligibility Predictions	Projects valid attempt windows beyond current year with a downloadable roadmap.	Timeline	P1 - High
Full Eligibility Report	Downloadable PDF of eligibility standing across all tracked exams.	Eligibility	P1 - High
Settings Screen	Profile info, notifications, security & privacy, dark mode, language preferences.	Settings	P1 - High
Help & Support Screen	FAQ browser with search, category cards, and support status banner.	Help	P2 - Future
Push Notifications	FCM-based alerts for deadlines and eligibility changes.	System	P2 - Future
Multi-language Support	Hindi and regional language UI strings via Flutter l10n.	Global	P2 - Future

6. App Flow / User Journey

6.1 First-Time User Journey

- User downloads and launches Yogya. Splash Screen checks for a JWT token in secure storage. None found - routes to Login Screen.
- User registers via email/password or Google OAuth. JWT issued and stored in the hardware-backed keystore.
- Profile Screen opens: user enters Date of Birth, Gender, Social Category, State of Domicile, Educational Status, and Primary Exam Goal.
- A 'Complete Your Profile' nudge directs the user to the Documents screen to upload their 10th Marksheet (first compulsory document).
- User opens the OCR viewfinder, aligns the marksheet within the frame. The 4-layer pipeline processes the image in real time, showing a progress bar from 0 % to 100 % with step badges.

- Extracted data is shown for user verification. On confirmation, the raw image is purged from the device. The Eligibility Engine performs a first pass using the now-verified date of birth.
- User uploads 12th Marksheet and Graduation Certificate through the same OCR flow. With all three core documents verified, the Live Eligibility Pulse shows full match percentages and the Dashboard surfaces the first eligible exam's registration deadline.

6.2 Returning User Journey (Daily Use)

- Splash Screen detects a valid JWT - routes directly to Dashboard. Cold-start under 2 seconds.
- Dashboard shows current UPSC attempt count (e.g., 3 / 6), nearest deadline with countdown (e.g., 'CDS II 2024 Application - 12 Days Remaining'), and recent document verification card.
- User opens the Eligibility tab, views the Live Eligibility Pulse, and updates CGPA after semester results. The engine recomputes all exam matches in real time.
- User opens the Timeline tab. Sees UPSC CSE Prelims (ACTIVE, Eligible, Attempt #2 / 6), AFCAT (Ineligible - age exceeded by 2 months), and CDS II 2024 (Eligible with 'Register Now' button).
- User taps 'Register Now' on CDS II. An 'Eligible - Next Steps' card opens listing required documents (12th certificate, degree certificate, NCC C-certificate if applicable, medical fitness clearance) with links to the official UPSC portal.

6.3 Navigation Flow

Entry Point	Route	Destination
Cold start (no session)	Splash → Login	Login Screen
Cold start (active session)	Splash → Dashboard	Dashboard
Bottom Nav - Tab 1	Direct	Dashboard
Bottom Nav - Tab 2	Direct	Eligibility Timeline
Bottom Nav - Tab 3	Direct	Documents / OCR Scanner
Bottom Nav - Tab 4	Direct	Profile Screen
Hamburger icon	Slide-in	Side Drawer
Drawer - Eligibility Engine	Push	Rule-Based Eligibility Wizard
Drawer - My Documents	Push	Documents Library View

Entry Point	Route	Destination
Drawer - Attempt History	Push	Attempt History Screen
Drawer - Settings	Push	Settings Screen
Eligible card - Register Now	External / WebView	Official Exam Portal
Sign Out (Settings)	go_router redirect	Login Screen

7. UI/UX Design - Wireframes & Visual Design

7.1 Design Language

Token	Value	Usage
Primary Colour	#3B5BDB (Indigo)	Buttons, active tab, brand wordmark, header gradients
Dark Background	#1A1A2E	Splash gradient bottom, dark motivational cards
Surface	#FFFFFF	Cards, input fields, drawer body
Page Background	#EEF1F8	Login screen and light page backgrounds
Eligible Green	#2ECC71	ELIGIBLE badges, verified ticks
Ineligible Red	#E74C3C	INELIGIBLE badges, error chips
Partial Amber	#F39C12	PARTIAL badges, deadline alert banners, OCR step dots
Border Radius	12–24 dp	All cards and input fields
Elevation	2–4 dp	Cards; 0 dp for flat list items

7.2 Screen-by-Screen Visual Design & Wireframes

7.2.0 Splash Screen

- Full-screen indigo gradient (top: #3B5BDB → bottom: #1A1A2E).

- ‘Yogya’ wordmark in white extra-bold, centred. Tagline ‘The Digital Sanctuary’ in light-weight text below.
- Logo appears with ScaleTransition + FadeTransition (800 ms, Curves.easeOutBack). Three dot indicators at the bottom show loading state.
- Minimum 2-second display; routes to Login (no session) or Dashboard (active session) on completion.

Figure 1 Spash Screen

7.2.1 Login Screen

- Light grey-blue background (#EEF1F8). Centred white card, 24 dp rounded corners, soft drop shadow.
- ‘Yogya’ is the brand name in indigo at the top of the card. ‘Welcome back’ in large bold black; subtitle in regular grey.
- Email field with mail icon prefix; Password field with lock icon and ‘Forgot Password?’ link.
- Full-width indigo gradient ‘Sign in →’ button. OR divider. ‘Continue with Google’ outlined button. ‘Sign Up’ redirect link.

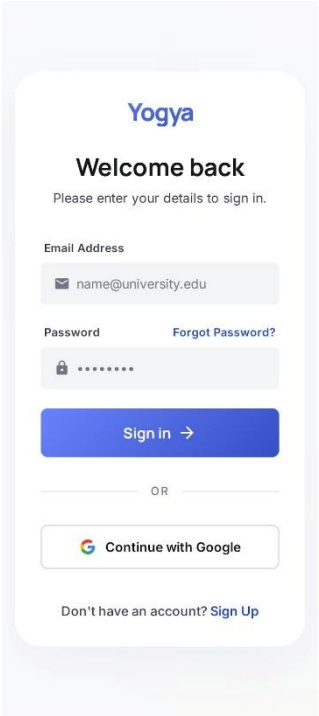

Figure 2 Login Screen

7.2.2 Dashboard Screen

- App bar: ‘Yogya’ in bold with hamburger icon (left) and notification bell (right).
- Eligibility Alert Banner: Pastel-gold strip - ‘You are eligible for upcoming CDS II 2024 - Check Details’.
- Current Standing Donut Card: White elevated card with a circular arc indicator (e.g., 3 / 6 Attempts Left), exam name and category below.

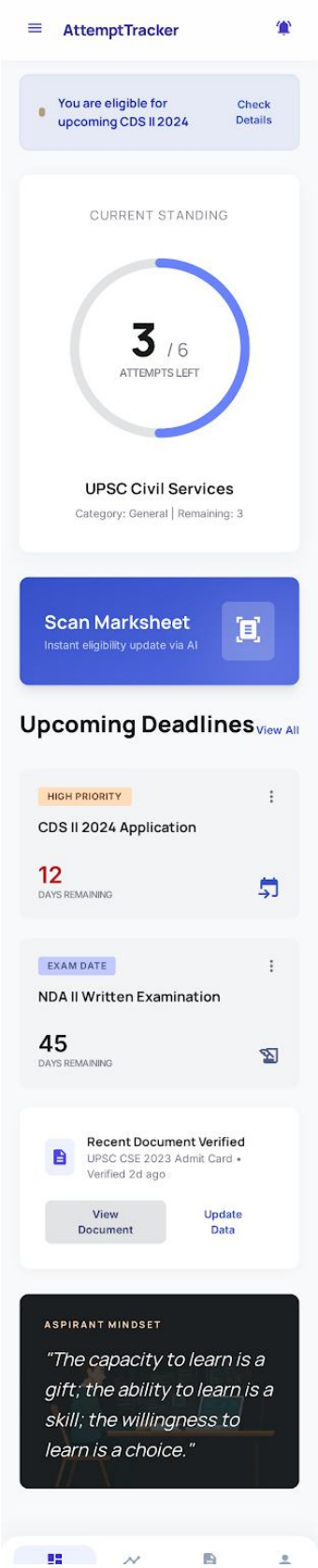

Figure 3 Dashboard

- Scan Marksheet CTA: Full-width indigo gradient card with scan icon. Tapping opens the OCR viewfinder.
- Upcoming Deadlines: Cards with HIGH PRIORITY / EXAM DATE badges and days remaining in large text.
- Recent Document Verified card with 'View Document' and 'Update Data' action buttons.
- Aspirant Mindset Quote: Dark indigo card with italic motivational quote, refreshes daily.

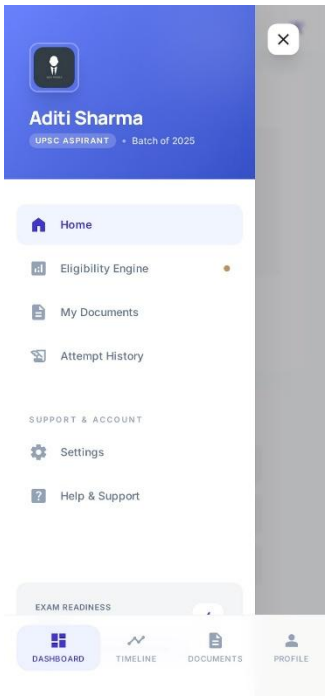

Figure 4 Side Drawer

7.2.3 Side Drawer

- Indigo gradient header: user avatar, full name, role tag (UPSC ASPIRANT), batch year.
- Navigation items: Home (highlighted), Eligibility Engine (amber dot badge), My Documents, Attempt History.
- Support & Account group: Settings, Help & Support.
- Partially visible Exam Readiness preview card at the bottom prompts profile completion.

7.2.4 Eligibility Engine Screen

- Page title: 'Rule-Based Eligibility' with subtitle describing the precision logic engine.
- Step 1 - Personal Identity: Social Category pill buttons (General | OBC | SC | ST), Date of Birth picker with note that age is calculated relative to each exam's notification date.
- Step 2 - Academic Records: Unlocks after Step 1. Fields for 10th %, 12th %, graduation CGPA / %, stream. Pre-filled from verified OCR documents where available.
- Step 3 - Review & Confirm: Summary of all entered data for confirmation before the engine runs.
- Save Draft and Next Phase buttons below the wizard steps.

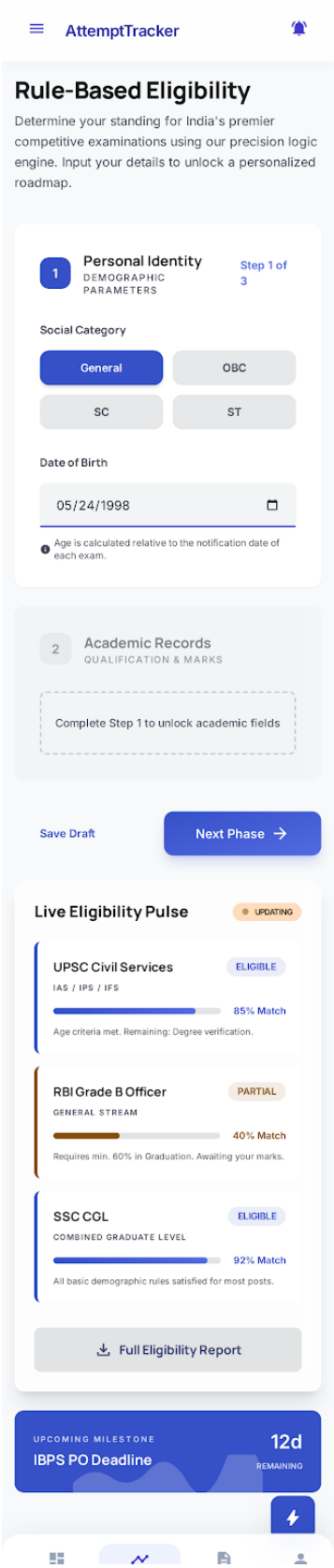

Figure 5 Eligibility Engine Screen

- Live Eligibility Pulse below: per-exam ELIGIBLE / PARTIAL / INELIGIBLE badges with coloured progress bars and one-line reason text.
- Full Eligibility Report download button at the bottom. IBPS PO Deadline milestone banner pinned at the foot of the screen.

7.2.5 Eligibility Timeline Screen

- Page title: 'Eligibility Timeline' with profile-personalised subtitle.
- Next Deadline Banner (indigo bordered card): deadline name, days remaining, REGISTER NOW button linking to the official portal.
- Timeline nodes drawn by the timeline_tile package: UPSC CSE Prelims (ACTIVE, Eligible, Attempt #2 / 6), AFCAT (Ineligible + red reason chip - age limit exceeded), CDS II 2024 (Eligible).
- Tapping a node expands it to show Age Limit (current vs. allowed), Qualification (Verified / Pending), and a View Details link.
- Future Eligibility Predictions card with DOWNLOAD ROADMAP action.
- Strategic Overview dark card: Current Age, Total Valid Attempts, personalised strategic message.
- Age Relief Claims amber-bordered card. Floating + FAB to add a new exam to tracking.

7.2.6 Documents & OCR Scanner Screen

- App bar: back arrow, Help (?), and notification bell icons.

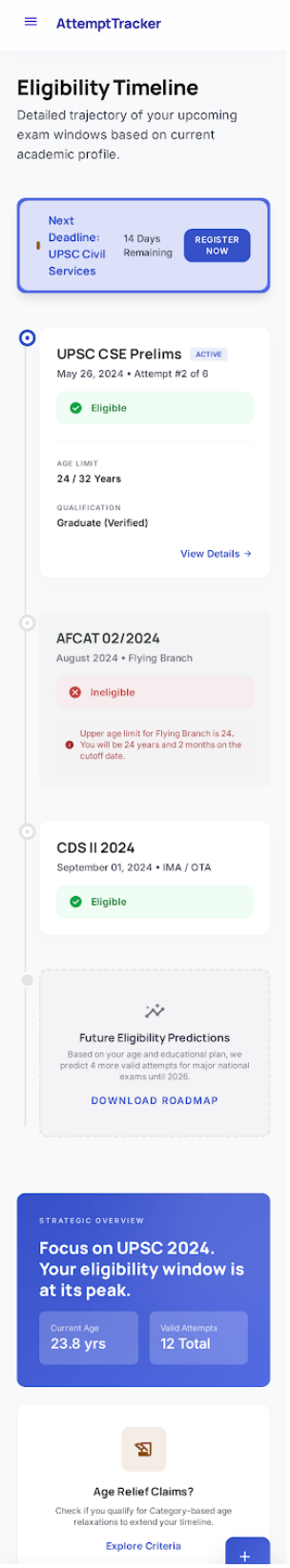

Figure 6 Eligibility Timeline Screen

Figure 7 Documents & OCR Scanner Screen

- OCR Viewfinder: Live camera preview with blue corner accents, animated scan line, 'ALIGN DOCUMENT WITHIN FRAME' guide badge, and Flash / Shutter / Gallery FABs.
- Progress Feedback Card: 'Extracting Marks Data... Processing Layer 2 of 4', percentage counter (e.g., 68 %), indigo-to-pink gradient bar, processing step badges (Table Detected, Text Alignment...).
- Recently Scanned section: PRIMARY RECORD card showing the 12th Grade Marksheet (Aggregate 88.4 %, Confidence 99.2 %), Verify Data button, and Share icon.
- Compact list below for other scanned records (e.g., JEE Advanced Rank Card, IELTS Score Certificate).
- Missing a document? dark indigo CTA card with 'Import Files' button for bulk PDF upload.

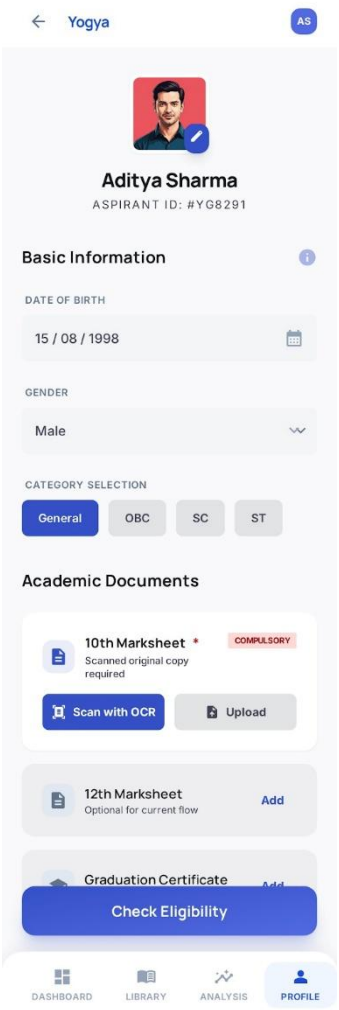

Figure 8 Profile Screen

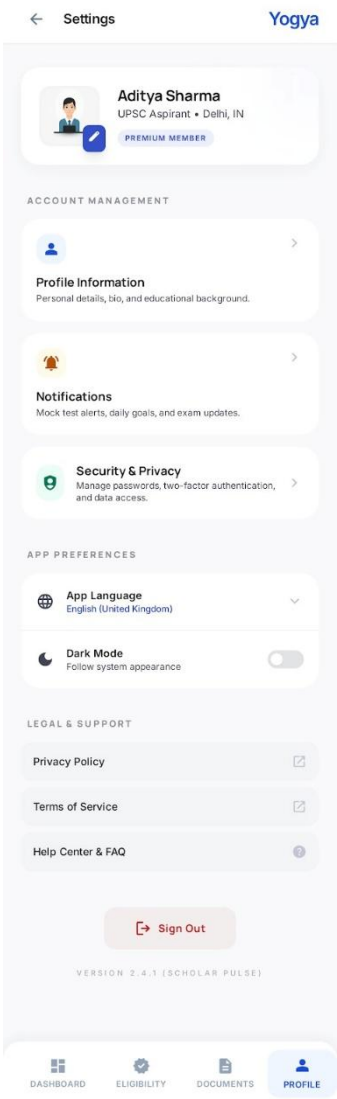

Figure 9 Settings Screen

7.2.7 Profile Screen

- Back arrow with 'Yogya' title; avatar initials button (top right). Centred avatar with edit pencil, full name, Aspirant ID.
- Basic Information section: Date of Birth (date picker), Gender (dropdown).
- Category Selection pill buttons: General | OBC | SC | ST (active state = indigo filled).
- Academic Documents section: 10th Marksheet (COMPULSORY badge, 'Scan with OCR' + 'Upload' buttons), 12th Marksheet (Add), Graduation Certificate (Add).
- Check the Eligibility primary action button at the bottom of the screen.

7.2.8 Settings Screen

- Back arrow, 'Settings' title, 'Yogya' brand (top right).
- Profile card at top: avatar, name, role (UPSC Aspirant, Delhi IN), PREMIUM MEMBER badge.

- Account Management: Profile Information, Notifications, Security & Privacy (each with a chevron arrow).
- App Preferences: App Language (dropdown, default English UK), Dark Mode (toggle, follows system).
- Legal & Support: Privacy Policy, Terms of Service, Help Center & FAQ.
- Red-tinted Sign Out button. Version string (VERSION 2.4.1 SCHOLAR PULSE) displayed below.

7.2.9 Help & Support Screen (Future Implementation)

- App bar: back arrow, 'Help & Support' title, 'Yogya' brand.
- Header banner (indigo rounded card): 'How can we help?' title, subtitle, white 'Search FAQs...' search bar.
- Browse Categories section with 'SELECTED TOPICS' label.
  - Exam Eligibility - ESSENTIAL badge; sub-topics: Age Criteria, Syllabus.
  - Account Help - sub-topics: Passwords, Security, Profile.
  - OCR Troubleshooting - sub-topics: Document Scan, Photo ID.
  - Documents & Certificates - sub-topics: Upload, Verify, Store.
- SUPPORT STATUS: HIGH VOLUME amber banner at bottom with animated pulse dots.

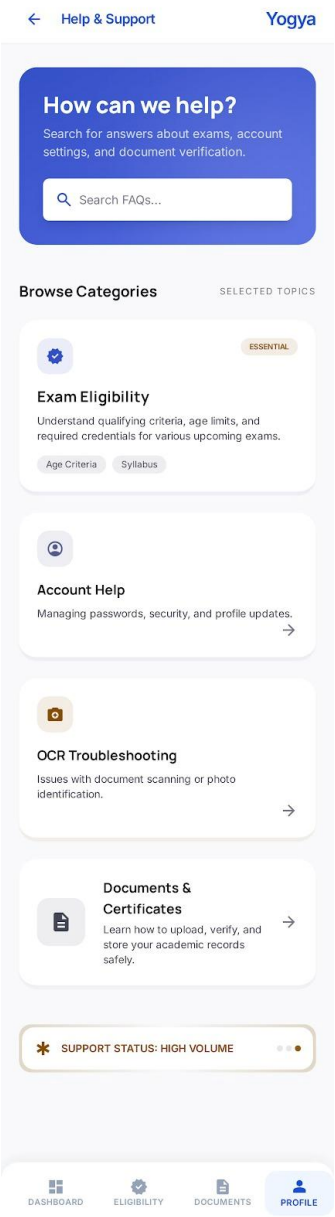

Figure 10 Help & Support Screen

8. Technical Architecture

8.1 Clean Architecture Layers

Layer	Responsibility	Key Contents
Presentation	UI rendering, state observation, user interaction.	Flutter Widgets, Riverpod Providers, go_router routes, Screen files.
Domain	Business rules and eligibility logic. Zero Flutter dependency.	EligibilityEngine, UseCases (ScanDocument, CheckEligibility), Entity models.

Layer	Responsibility	Key Contents
Data	API communication, local DB, OCR, secure storage.	Dio HTTP client, Hive / Isar local DB, ML Kit / TFLite OCR, flutter_secure_storage.

8.2 State Management - Riverpod 3.0

State Domain	Provider Type	Description
Authentication (JWT)	StateNotifierProvider	Manages login / logout state; persists token via flutter_secure_storage.
Camera / Flash	StateNotifierProvider	Flash ON / OFF, front / back toggle for the OCR viewfinder.
OCR Progress	AsyncNotifier	Yields continuous % updates (0→100) driving animated progress bars without full rebuilds.
Eligibility Engine	EligibilityNotifier	Recomputes match % for all tracked exams on every profile or document change.
Document Library	FutureProvider	Fetches and caches scanned document records from local DB + remote API.
Timeline Events	FutureProvider	Fetches and caches exam timeline data; supports offline display.
Profile / Category	StateNotifierProvider	Manages social category selection; propagates to EligibilityNotifier on every change.

8.3 OCR Pipeline - 4 Layers

- Layer 1 - Image Pre-processing: Perspective correction, deskewing, contrast enhancement, and noise reduction using OpenCV bindings or Flutter's image package.
- Layer 2 - Layout Detection: Identifies document type (marksheet vs. degree), detects table regions and column headers using a lightweight on-device TFLite model.
- Layer 3 - Text Extraction: Runs Google ML Kit Text Recognition on detected regions to extract raw text strings.
- Layer 4 - Structured Data Parsing: A rule-based parser maps text to a typed schema (board, year, marks, aggregate %, DOB). Per-field confidence scores calculated. Raw image purged via dart:io File.delete() on success.

9. Data Design

9.1 Local Database Schema (Hive / Isar)

Collection	Key Fields	Notes
UserProfile	userId, name, dateOfBirth, gender, socialCategory, stateOfDomicile, educationalStatus, primaryExamGoal	Social category stored encrypted. No certificate document ever stored.
AcademicDocument	docId, docType, boardName, year, aggregate, cgpa, stream, confidence, verifiedAt	Structured fields only. Raw image never persisted.
EligibilityResult	resultId, examId, matchPercent, status, reasonText, computedAt	Cached per exam; recomputed on profile or document change.
TrackedExam	examId, examName, notificationDate, registrationDeadline, registrationUrl, additionalDocsRequired	Populated from remote API; cached for offline access.
AttemptRecord	attemptId, examId, examYear, attemptNumber, documentRef	Derived from uploaded admit cards or manual entry.
DeadlineAlert	alertId, examId, deadlineDays, priority, snoozedUntil	Drives Dashboard countdown cards.

9.2 Data Flow

- User scans document → OCR pipeline extracts structured fields → AcademicDocument record written to local DB.
- AcademicDocument write triggers EligibilityNotifier → EligibilityEngine reads UserProfile + all AcademicDocuments → computes new EligibilityResult per TrackedExam.
- Dashboard and Timeline observe EligibilityResult stream via Riverpod FutureProvider → only the changed card rebuilds.
- On an app open with an active session, TrackedExam and DeadlineAlert data is refreshed from the remote API using a stale-while-revalidate caching strategy.

9.3 Privacy & Data Minimisation

- Raw document images retained only in volatile memory (RAM) during OCR processing. Purged immediately on extraction completion.
- Social category stored as an encrypted string in UserProfile. No physical certificate document is ever stored or transmitted.
- JWT tokens stored exclusively in flutter_secure_storage (iOS Keychain / Android Keystore). Never in SharedPreferences or any plaintext file.
- Only structured EligibilityResult and UserProfile fields are synced to the backend. Document images and raw OCR text are never transmitted.

10. APIs & Integrations

10.1 Internal REST API

Endpoint	Method	Purpose	Auth
POST /auth/login	POST	Email / password authentication; returns Yogya JWT.	No
POST /auth/google	POST	Exchanges Google ID token for Yogya JWT.	No
GET /exams	GET	Returns all TrackedExams with eligibility rules, deadlines, and registration URLs.	Yes
POST /profile	POST	Creates or updates UserProfile on the backend.	Yes
POST /documents	POST	Syncs structured AcademicDocument fields (no images) to backend.	Yes
GET /eligibility/report	GET	Generates Full Eligibility Report PDF server-side; returns download URL.	Yes
GET /deadlines	GET	Returns upcoming deadlines ordered by urgency for the user's profile.	Yes

10.2 Third-Party Packages

Package	Purpose
google_sign_in	OAuth 2.0 authentication via Google account.
google_mlkit_text_recognition	On-device OCR for document text extraction.
tflite_flutter	Layout detection model for document type classification.
flutter_secure_storage	Hardware-backed JWT and credential storage.

Package	Purpose
mobile_scanner / camera	Live document capture for the OCR viewfinder.
image_picker	Gallery-based document import for bulk upload.
dio	REST API communication with JWT interceptors and retry logic.
hive / isar	Offline-first embedded local database.
go_router	Declarative navigation, deep linking, and auth guards.
flutter_riverpod 3.0	Global and scoped state management across all screens.
freezed + json_serializable	Type-safe DTOs and immutable data models.
fl_chart	Dashboard donut chart and eligibility match progress bars.
timeline_tile	Eligibility Timeline node rendering.
firebase_messaging	Future (P2): push notifications for deadline alerts.

10.3 Official Exam Portal Links

Exam	Registration Portal	Exam Info Link
UPSC Civil Services (IAS / IPS / IFS)	upsconline.nic.in	upsc.gov.in/examinations/civil-services-examination
UPSC CDS (IMA / OTA / AFA)	upsconline.nic.in	upsc.gov.in/examinations/combined-defence-services-examination
UPSC NDA & NA	upsconline.nic.in	upsc.gov.in/examinations/national-defence-academy-and-naval-academy-examination
AFCAT	afcat.cdac.in	indianairforce.nic.in/content/afcat
SSC CGL	ssc.gov.in	ssc.gov.in (notifications section)
SSC CHSL	ssc.gov.in	ssc.gov.in (notifications section)
IBPS PO	ibps.in	ibps.in/crp-po-mt.html
IBPS Clerk	ibps.in	ibps.in/crp-clerk.html
RBI Grade B Officer	opportunities.rbi.org.in	rbi.org.in/Scripts/Vacancies.aspx
State PSC (dynamic)	Resolved from user's State of Domicile in Profile	Linked dynamically in-app

Note: Yogya does not store or upload these additional documents. It surfaces each requirement as an informational card with a direct link so the user can submit documents on the official portal at the time of registration.

11. Development Plan

11.1 Technology Stack

Layer	Technology	Rationale
Frontend Framework	Flutter (Dart)	Single codebase for Android & iOS; excellent animation support.
State Management	Riverpod 3.0	Compile-safe, testable, no BuildContext dependency.
Routing	go_router	Declarative, supports deep links and auth guards natively.
Networking	Dio	Interceptors, retry logic, multipart support.
Local DB	Hive / Isar	Fast, NoSQL, Flutter-native; supports offline-first.
OCR	ML Kit + TFLite	On-device; no image upload required; preserves privacy.
Auth Storage	flutter_secure_storage	Hardware-encrypted credential storage (Keychain / Keystore).
Code Generation	freezed + json_serializable	Reduces boilerplate; eliminates JSON parsing errors.
Backend	Node.js / Firebase (TBD)	Lightweight REST API for user data sync and report generation.

11.2 Sprint Plan

Sprint	Duration	Deliverables
Sprint 0 - Setup	1 week	Project scaffold, Flutter environment, go_router setup, Riverpod skeleton, CI/CD pipeline, design token implementation.
Sprint 1 - Auth & Profile	2 weeks	Splash Screen, Login Screen (email + Google OAuth), Profile Screen (Basic Info, Category, Academic Docs section), flutter_secure_storage integration.
Sprint 2 - OCR Pipeline	3 weeks	Camera viewfinder, 4-layer OCR pipeline, progress feedback card, 10th Marksheet extraction, document verification UI, data purge logic.
Sprint 3 - Core Documents	2 weeks	12th Marksheet OCR, Graduation CGPA OCR, auto-CGPA-to-% conversion, document library UI, bulk import.
Sprint 4 - Eligibility Engine	3 weeks	Rule-based eligibility engine for 10 exams, Live Eligibility Pulse, 3-step wizard UI, EligibilityNotifier, real-time updates.

Sprint	Duration	Deliverables
Sprint 5 - Dashboard & Timeline	2 weeks	Dashboard (donut, deadlines, quote), Eligibility Timeline (timeline_tile nodes), Next Steps cards, registration links.
Sprint 6 - Navigation & Settings	1 week	Side drawer, Attempt History screen, Settings screen, Sign Out, Help & Support skeleton.
Sprint 7 - API Integration	2 weeks	Backend API integration, TrackedExam sync, deadline refresh, Full Eligibility Report PDF generation.
Sprint 8 - QA & Polish	2 weeks	End-to-end testing, performance profiling, accessibility review, animation polish, store submission preparation.
Sprint 9 - v2 Features	Ongoing	Push notifications (FCM), multi-language support (Flutter l10n), Help & Support full build, Password Reset flow.

11.3 Team Responsibilities

Team Member	Primary Responsibility
Aditya Kumar Yadav	OCR pipeline implementation (ML Kit + TFLite), document processing, image pre-processing.
Gaurav Yadav	Architecture design, Riverpod state management, Eligibility Engine business logic, API integration.
Hitesh Purohit	UI implementation (Dashboard, Timeline, Settings, Help screens), animation, go_router navigation.
Aman	Profile screen, Login screen, secure storage, backend API design, database schema implementation.

12. Deliverables

12.1 Software Deliverables

- Fully functional Flutter application (Android APK + iOS IPA) covering all screens specified in Sections 7 and 8.
- OCR pipeline processing 10th Marksheet, 12th Marksheet, and Graduation CGPA / Degree Certificate at ≥ 95 % and ≥ 90 % confidence thresholds respectively.
- Rule-based Eligibility Engine covering UPSC CSE, CDS, NDA, AFCAT, SSC CGL, SSC CHSL, IBPS PO, IBPS Clerk, RBI Grade B, and State PSC examinations.
- Secure authentication system (Email / Password + Google OAuth) with JWT stored in hardware-backed secure storage.

- Offline-first local database with structured academic data, eligibility results, and deadline alerts.
- Integration with official exam portal URLs for registration links and additional document information.

12.2 Design Deliverables

- This Design Document covering all 12 sections: Problem Statement, Objectives, User Personas, Competitive Analysis, Features List, App Flow, UI/UX Design, Technical Architecture, Data Design, APIs & Integrations, Development Plan, and Deliverables.
- Screen-level wireframes embedded in Section 7.2 with reference UI screenshots from the Yogya prototype.
- Design token specification (colours, typography, border radii, elevation) documented in Section 7.1.

12.3 Documentation Deliverables

- API documentation for all internal REST endpoints (Section 10.1).
- Data schema documentation for all local database collections (Section 9.1).
- Third-party integration list with package names and purposes (Section 10.2).
- Sprint plan and team responsibility matrix (Sections 11.2 and 11.3).

12.4 Quality Assurance Deliverables

- Unit tests for the Eligibility Engine covering all 10 tracked examinations and all category variants (General, OBC, SC, ST, EWS).
- Widget tests for the OCR progress feedback card, Live Eligibility Pulse, and Timeline nodes.
- Integration test for the end-to-end flow: document scan → OCR extraction → eligibility computation → dashboard update.
- Performance benchmark: app cold-start under 2 seconds; OCR processing under 5 seconds for a standard A4 marksheet photograph.

- End of Document -